

Mohammad Mahmudul Hasan

Project Lead, AI Department · HawarIT Limited

Dhaka, Bangladesh
+8801521434403
mahmudul18@iut-dhaka.edu
fahome10bd.github.io
GitHub: fahome10bd
LinkedIn: fahome10bd

Research Interests

Geospatial AI; 3D scene understanding and point cloud analytics; urban reconstruction and digital twins; autonomous GIS workflows and spatiotemporal change detection.

Education

2017-2021 B.Sc. in Electrical and Electronic Engineering

Islamic University of Technology (IUT), Bangladesh.

CGPA: **3.76/4.00**, first class honours.

Thesis *Quantifying Locomotive Features in EEG of Impaired Consciousness (Coma) with Distinctive Cerebral Rhythms*. Analyzed EEG time series and cerebral rhythms. A+ in Project and Thesis courses. IUT repository.

Peer-Reviewed Publication

2021 Tasnim Sakib Apon, **Mohammad Mahmudul Hasan**, Abrar Islam, and Md. Golam Rabiul Alam. "Demystifying Deep Learning Models for Retinal OCT Disease Classification using Explainable AI." *2021 IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE)*.
DOI: 10.1109/CSDE53843.2021.9718400 · Author preprint.

Professional Experience

2024-Present **Project Lead, AI Department** · HawarIT Limited.
Lead multidisciplinary teams delivering AI and geospatial automation systems.

2023-2024 **Senior AI Engineer** · HawarIT Limited.
Developed AI systems for photogrammetry, point cloud analytics, and spatial automation.

2021-2023 **ML Engineer** · HawarIT Limited.
Built machine learning and computer vision pipelines for geospatial applications.

Selected Technical and Research Projects

Point clouds	Developed LiDAR semantic segmentation and automated pipe/geometry extraction across aerial, mobile, and indoor datasets: 95% extraction accuracy and 60% less manual processing time.
Change detection	Developed multi-temporal building change detection and integrated GIS validation: 88% accuracy and 65% less manual inspection.
Orthophoto QA	Automated cloud/shadow/blur assessment (97% accuracy), PostGIS/AHN seamline refinement (65% less editing), and camera alignment (2-pixel accuracy; 90% less manual aerial triangulation effort).
Street-view GeoAI	Developed YOLOv5 and tracking pipelines for 130 traffic sign and 33 street furniture classes: detection accuracy above 95% and localization precision of 92%.
3D city models	Built CityGML reconstruction and occlusion-aware multi-view aerial texture projection, maintaining reported texture displacement below 3 pixels.
Remote sensing	Applied Sentinel multispectral analytics to crop classification, water quality, and algae bloom detection. The framework placed second in an international innovation tender.

Technical Skills

Programming	Python, SQL, JavaScript, C, MATLAB, R.
AI and vision	PyTorch, TensorFlow, OpenCV, YOLO, FastAPI.
Geospatial	GDAL, GeoPandas, PostgreSQL/PostGIS, QGIS, ArcGIS.
3D and LiDAR	Open3D, PCL, COLMAP, OpenMVS, CloudCompare.

Awards and Honors

2024 / 2022	Excellent Performer (2024); Borsho Shera, Best Performer of the Year (2022), HawarIT Limited.
Selected awards	2nd place, International Innovation Tender; 1st place, MATLAB Competition, AUST; Top 15, Dhaka.AI Deep Learning Competition; 1st prize, Nuclear Power Seminar, AUST; Government Merit Scholarship, HSC.

References

Mirza Fuad Adnan Assistant Professor Dept. of EEE, IUT adnan152616@gmail.com	Md. Thesun Al-Amin Assistant Professor Dept. of EEE, IUT thesun.eee@gmail.com	Kaisar Imam Chief Technology Officer HawarIT Limited k.imam@hawarIT.com
---	---	---